

Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

Xiaofan Liu^{1*†}, Ronghao Zhang^{1†} and Yuan Feng^{1†}

^{1*}School of Software Engineering, East China Normal University, Shanghai, China.

*Corresponding author(s). E-mail(s): 10244602411@stu.ecnu.edu.cn;

[†]These authors contributed equally to this work.

Abstract

Purpose Calligraphy practice systems should expose local stroke structure while tolerating harmless placement variation. We study this problem in a low-resource direct-digital-writing setting.

Methods We represent each character by five overlapping direction masks and one keypoint mask. U-Net, DeepLabV3+, and SegFormer-B2 are evaluated with three seeds on quality-controlled standard and character-disjoint splits. A constrained similarity registration is assessed on 200 real references by controlled perturbations and paired ablations. We introduce aligned spatial-distribution similarity (ASDS), which compares polar occupancy, coarse spatial occupancy, and projection profiles after registration. Score validity is studied with three blinded raters on 150 natural same-character pairs. ASDS is developed retrospectively and evaluated by character-grouped out-of-fold internal validation. A paired direct-ink ablation tests whether its scalar result depends on parsing.

Results DeepLabV3+ obtains 0.9630 ± 0.0006 standard-split Macro Dice, whereas SegFormer-B2 leads character-disjoint evaluation at 0.8866 ± 0.0008 . Constrained registration reduces nuisance penalties by 30.118 points (95% CI 29.506–30.727) relative to no alignment while preserving an additional 11.189 points of structural penalty (10.094–12.322). The frozen production and coverage-aware scores correlate with blinded ratings at $\rho = 0.297$ and 0.429 , respectively. In retrospective analysis, ASDS reaches $\rho = 0.556$, with character-grouped out-of-fold $\rho = 0.540$. Parsed-mask and direct-ink ASDS are nearly equivalent ($\Delta\rho = 0.0004$, 95% CI $-0.0101-0.0103$).

Conclusion Overlapping parsing and constrained registration provide auditable structural evidence, while spatial-distribution comparison improves internal human association. ASDS remains retrospective and requires external replication; the scores do not represent universal calligraphic beauty.

Keywords: Chinese calligraphy, document analysis, multi-label segmentation, reference-based assessment, human validation, explainable feedback

1 Introduction

Digital tools for Chinese calligraphy usually solve one of three tasks: recognizing the written character, identifying a calligraphic style, or generating

a visually plausible glyph. A practice system has a different requirement. Given a learner’s writing and a designated reference, it should show which structural regions agree, which regions are missing

or excessive, and how the global organization differs. The output must be local enough to support revision, yet conservative enough not to masquerade as an expert judgment of brushwork, rhythm, historical authenticity, or beauty.

This requirement creates a document-analysis problem at the interface of segmentation, registration, shape comparison, and human-centred evaluation. First, strokes in a finished raster character intersect and adhere. A pixel at an intersection can legitimately belong to more than one directional structure, making mutually exclusive semantic segmentation inappropriate. Second, direct overlap confounds structural disagreement with global translation, scale, and small rotation. Conversely, an unconstrained warp may erase the very local discrepancy that a teaching system should report. Third, a mathematically convenient overlap score need not follow human perception. The score therefore requires explicit semantic auditing and human validation, not only a high self-consistency result.

The low-resource regime makes the problem harder. Our recovered corpus contains complete six-channel ground truth for only 40 character identities. Writer identity and style provenance cannot be recovered reliably. In this setting, a strong standard-split result can overstate generalization because the same character identities appear during training and testing. We therefore separate ordinary quality-controlled evaluation from a frozen character-disjoint protocol. This distinction follows the broader document analysis concern that performance under limited labels and unseen writing conditions must be measured rather than inferred from an in-domain split [1, 16].

We study the following research questions.

RQ1: How accurately do convolutional and transformer parsers recover the overlapping direction and keypoint representation, and how much performance is lost on unseen character identities?

RQ2: Does a deliberately constrained global registration reduce nuisance sensitivity relative to no alignment while retaining sensitivity to local structural changes?

RQ3: How well do fixed reference-conditioned scores agree with blinded human structural-similarity ratings, and can an interpretable spatial-distribution score improve that association

without an unacknowledged post-hoc claim? Does its scalar association depend on parser-derived union ink rather than direct source ink?

RQ4: Which diagnostic evidence can be localized reliably, and where do the current rules or evaluation protocol fail?

Our pipeline separates perception, comparison, and presentation:

overlapping parsing $\rightarrow$ constrained registration
 $\rightarrow$ structural comparison
 $\rightarrow$ localized evidence
 $\rightarrow$ text realization.

The parser produces five independently activated direction channels and one keypoint channel. The deployed comparison retains a frozen Dice/IoU/keypoint formula. The paper additionally develops aligned spatial-distribution similarity (ASDS), a compact score based on Jensen–Shannon similarity between polar occupancy, coarse grid occupancy, and horizontal/vertical projection profiles. ASDS is intentionally low-capacity and uses neither character identity nor human ratings at inference. Because its design followed the original rating study, we report development and character-grouped out-of-fold internal-validation results separately and preserve the complete candidate search rather than presenting the apparent optimum alone.

The contributions are:

1. an overlapping six-channel representation and reproducible quality-control protocol for complete-character stroke-structure parsing;
2. a three-architecture, three-seed comparison on both standard and character-disjoint splits, exposing an architecture-dependent trade-off between dense direction masks and sparse keypoints;
3. a constrained registration method evaluated by real-reference controlled perturbations, paired bootstrap intervals, signed-rank tests, and a wider-range negative ablation;
4. an audit of inactive-channel credit and a blinded three-rater study on 150 natural same-character, different-instance pairs;
5. ASDS, an interpretable spatial comparison whose post-rating development status, grouped cross-validation, component ablations, and

remaining need for external replication are reported explicitly, together with a paired direct-ink ablation that separates scalar silhouette scoring from the parser’s direction- and endpoint-level diagnostic role;

6. a diagnostic failure taxonomy that distinguishes scoring limitations, localization-metric limitations, and optional language-model realization.

The intended input is a raster exported from a direct digital writing canvas, not a photograph of paper. Accordingly, the claims concern structural analysis in this modality. They do not cover camera robustness, writer- disjoint generalization, stroke order recovery, or universal aesthetic grading.

2 Related Work

2.1 Chinese handwriting recognition and document segmentation

Large databases such as CASIA-HWDB established modern benchmarks for online and offline handwritten Chinese character recognition [13]. Recognition assigns a character identity, whereas our task requires spatially resolved evidence about directional stroke regions and keypoints. It is therefore closer to document segmentation than to closed-set classification.

U-Net introduced an encoder–decoder architecture with skip connections [18]; DeepLabV3+ combines atrous spatial pyramids with a decoder [3]; and SegFormer uses hierarchical transformer features with a lightweight multilayer-perceptron decoder [23]. Generic document models such as dhSegment illustrate the value of reusable segmentation formulations [16]. Recent IJDAR work further shows that label scarcity, source-target mismatch, and the choice of pretraining protocol can materially affect handwritten document segmentation [1]. We use established architectures to study a different output contract: the five direction labels are independently activated because intersections can support more than one label.

Stroke-Seg is the most directly related recent framework for Chinese character stroke segmentation [8]. It combines a synthetic-image generation strategy with interchangeable semantic segmentation backbones to separate character strokes. Our

representation addresses a different annotation and application contract. Each manually isolated stroke is assigned by a fixed, character-specific mapping to one of five direction families; crossings may therefore activate multiple channels at the same pixel, and a sixth mask records the union of annotated stroke endpoints. The objective is not to recover an ordered list of individual strokes. Instead, the representation supports overlapping structural parsing, reference-conditioned comparison, and direction-level diagnostic evidence.

2.2 Computational analysis of Chinese calligraphy

Computational calligraphy research spans handcrafted style descriptors, recognition, generation, and feature extraction. Earlier work represented calligraphic style using local and global shape features [17, 26]. CalliNet advances style classification with metric learning and triplet supervision [25]. Other systems extract artistic regions from a calligraphy work relative to a reference [19], beautify handwritten glyphs toward a Kai-style template [22], or evaluate robotic writing through designed aesthetic features [14].

Large contemporary resources broaden the available calligrapher, script, and page-level annotations. UniCalli, for example, supports column-level calligraphy generation and related representation learning [24]. These resources are valuable for style generation and recognition, but they do not remove the need to validate a practice score against human judgments. Our present study is deliberately narrower: it compares two instances of the same character against a selected reference and reports structural evidence rather than claiming author authentication or style generation.

2.3 Handwritten-character quality assessment

A recent survey divides handwritten Chinese character quality evaluation into rule-based geometric measures, template comparison, and learned methods, while emphasizing the difficulty of defining quality labels and explaining model decisions [4]. Pairwise and Siamese approaches can learn similarity from labelled examples [5]. Multidimensional calligraphy evaluation has also been studied using visual, quality, and aesthetic attributes

[20]. These approaches demonstrate that “quality” is not a single observable quantity: legibility, geometric correctness, reference similarity, brushwork, and artistic appeal can disagree.

Our score is consequently scoped as *reference-conditioned structural agreement*. The human study asks raters to judge same-character structural similarity, not beauty or historical fidelity. This distinction is central to the validity claim: a moderate association with structural ratings cannot be relabelled as expert aesthetic calibration.

2.4 Registration and shape comparison

Registration methods range from parametric geometric alignment to learned spatial transformers [9, 27]. Flexible registration is useful when deformation is a nuisance, but it is hazardous for diagnostic assessment: anisotropic or deformable warping can conceal a proportion or placement error. We therefore restrict alignment to translation, isotropic scale, and small rotation and evaluate that policy with both nuisance and structural perturbations.

Shape contexts describe the spatial distribution of contour points in log-polar bins [2]; spatial pyramids retain coarse layout that an orderless representation would discard [11]. ASDS follows the same general principle but uses binary-ink occupancy rather than point correspondences. Its distributions are compared with the bounded, symmetric Jensen–Shannon divergence [12]. Polar occupancy captures relative radial/angular organization, a 3×3 grid retains coarse canvas layout, and projection profiles retain row/column mass. The design is simpler than a learned quality regressor and remains directly inspectable.

2.5 Human validation and diagnostic explanation

Automated assessment requires both association with human judgment and a clear account of rater reliability. We report two-way random-effects, absolute-agreement intraclass correlations following established ICC definitions and reporting guidance [10, 15]. Uncertainty for score correlations is estimated by bootstrap resampling at the character level [7]; paired alignment comparisons

additionally use the Wilcoxon signed-rank test [21].

Explanation is evaluated separately from score validity. The system first emits normalized evidence—direction channel, missing or extra structure, spatial region, centre offset, scale ratio, and keypoint agreement. A language model may verbalize this evidence, but it does not determine the score. This separation prevents fluent text from being treated as evidence of diagnostic correctness.

3 Method

3.1 Problem formulation

Let I_u be a learner image and I_r an approved reference image of the same target character. The system first predicts overlapping structural masks, then aligns the reference to the learner with a restricted global transform. It returns (i) six masks, (ii) a scalar structural-agreement score, and (iii) localized evidence (Fig. 1). Cross-character scoring is prohibited by the application contract.

3.2 Overlapping six-channel parsing

For an RGB character image I , a parser f_θ predicts

$$P = f_\theta(I) \in [0, 1]^{H \times W \times 6}.$$

The fixed channel order is

$$(\text{vec1}, \text{vec2}, \text{vec3}, \text{vec4}, \text{vec5}, \text{keypoint}).$$

The first five channels have the following fixed meanings: vec1 is vertical, vec2 is a rising diagonal (/), vec3 is horizontal, vec4 is a falling diagonal (\), and vec5 contains compound, curved, or otherwise non-cardinal strokes that cannot be assigned to the first four families. The sixth channel is the union of the two manually marked endpoints from each isolated stroke image. It does not encode junctions, turning points, or stroke order.

The ground-truth protocol begins with the completed character and its individually isolated stroke images. A fixed, character-specific `STROKE_VECTOR_MAP` assigns each isolated stroke

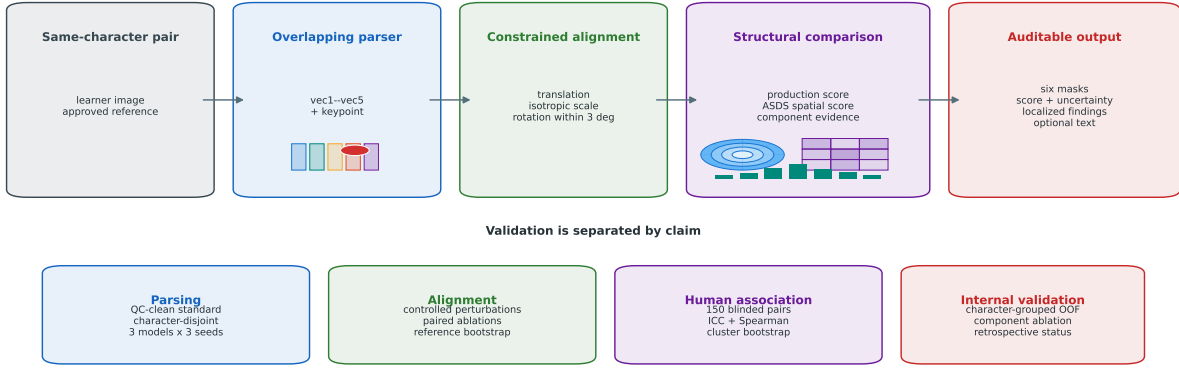


Fig. 1 System and validation overview. The reference and learner are parsed into overlapping direction/keypoint masks, globally aligned under a restricted transform, and compared by the frozen production score and ASDS. Segmentation, alignment, and human association are evaluated by separate protocols so that one successful experiment is not used to support a different claim.

to one of the five direction families; pixels from strokes assigned to the same family are merged. The endpoint annotations from all isolated strokes are merged into the keypoint mask. Because independently acquired strokes may occupy the same pixel after composition, a crossing can legitimately activate multiple direction channels. Each channel therefore uses an independent sigmoid rather than a mutually exclusive softmax (Fig. 2).

Images are converted to RGB, letterboxed to 512×512 , and normalized. Masks are resized by nearest-neighbour interpolation. For direction channel c , the loss combines class-weighted binary cross entropy and soft Dice:

$$\mathcal{L}_{\text{dir}} = \sum_{c=1}^5 \left(\mathcal{L}_{\text{BCE}}^{(c)} + \mathcal{L}_{\text{Dice}}^{(c)} \right).$$

The sparse keypoint target uses focal loss with $\gamma = 2$ and $\alpha = 0.75$, plus Dice loss. A boundary term derived from the masks is weighted by 0.2:

$$\mathcal{L} = \mathcal{L}_{\text{dir}} + \mathcal{L}_{\text{focal,kp}} + \mathcal{L}_{\text{Dice,kp}} + 0.2\mathcal{L}_{\text{boundary}}.$$

3.3 Versioned reference cache

Approved reference images are parsed once and stored as binary $512 \times 512 \times 6$ arrays. Each cache entry records its reference ID, character, course/style category, parser version, checkpoint SHA-256, thresholds, and channel schema. Caching removes repeated GPU inference for the

reference branch. These masks are model-derived references and are never treated as segmentation ground truth.

3.4 Constrained similarity registration

Let U and R denote the learner and reference binary stacks, and let

$$M(X) = \bigvee_{c=1}^5 X_c$$

be the union of direction ink. For each candidate isotropic scale s and rotation ϕ , the transformed reference is translated so that its foreground centroid approaches the learner centroid. The production search is bounded by

$$s \in [0.8, 1.2], \quad \phi \in [-3^\circ, 3^\circ],$$

and selects the candidate maximizing

$$J(M(U), M(T_{s,\phi,t}(R))),$$

where J is foreground IoU. Nonuniform scale, shear, thin-plate splines, and deformable warping are disallowed. This bounded search is a conservative compensation policy: it can reduce penalties from page placement and modest global size or rotation differences, while leaving local proportion and stroke-region errors available for diagnosis. It

Recovered six-channel annotation protocol (sample 33/18)

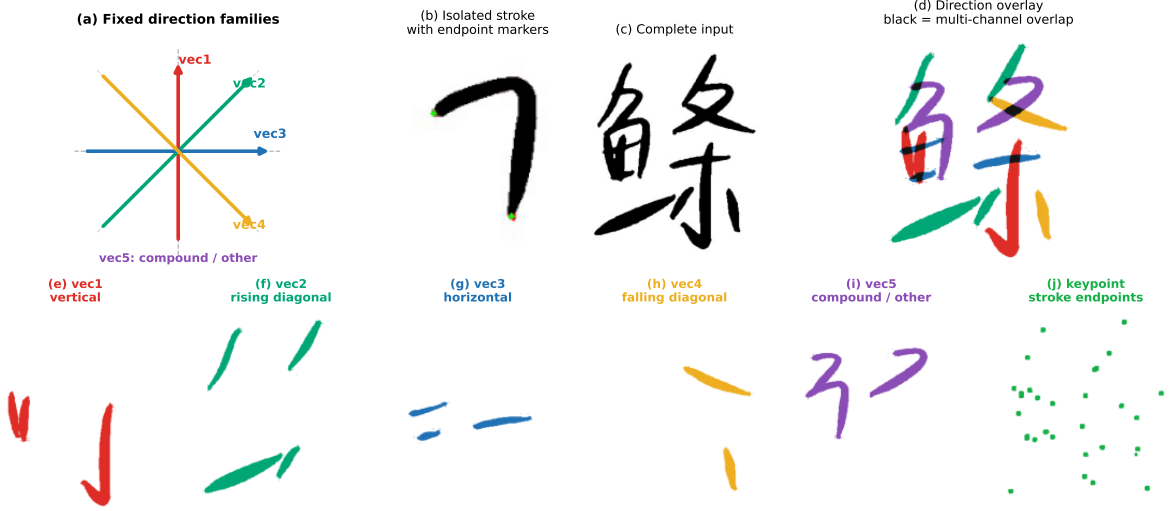


Fig. 2 Ground-truth six-channel definition. The direction guide defines vertical (vec1), rising-diagonal (vec2), horizontal (vec3), falling-diagonal (vec4), and compound/curved/other (vec5) families. Isolated stroke images are assigned through a fixed character-specific mapping and composed into overlapping masks; black in the colour overlay denotes pixels shared by more than one direction channel. The keypoint mask is the union of annotated stroke endpoints only. Endpoint markers are dilated in this figure for print visibility; the stored ground truth is unchanged.

does not imply invariance to all transforms within the nominal range.

3.5 Frozen production score

After alignment, let D_c be Dice agreement in direction channel c ,

$$\bar{D} = \frac{1}{5} \sum_{c=1}^5 D_c,$$

let J_{ink} be union-ink IoU, and let $F_{\text{kp}}^{(3)}$ be keypoint-mask F1 with a three-pixel tolerance. The deployed score is

$$S_{\text{prod}} = 100 \left(0.55\bar{D} + 0.25J_{\text{ink}} + 0.20F_{\text{kp}}^{(3)} \right).$$

This formula was frozen before the controlled experiments. An audit variant excludes direction channels that are empty in both images and avoids granting perfect keypoint credit when both keypoint masks are empty. The audit score is not retroactively substituted for S_{prod} .

3.6 Aligned spatial-distribution similarity

The human study showed that pixel- and channel-overlap terms incompletely capture perceived global organization. We therefore develop aligned spatial-distribution similarity (ASDS) on the union direction ink after the same frozen registration. The score uses three complementary distributions.

3.6.1 Polar occupancy

For a binary ink mask M , foreground coordinates are centred at their ink centroid. Radius is divided by the 95th percentile of foreground radius and clipped to $[0, 1]$. Each pixel is assigned to one of four radial and eight angular bins, producing a normalized 32-bin histogram $h_p(M)$. Independent centroid and radius normalization make this term resistant to small residual translation and isotropic scale, while retaining angular and radial mass organization.

3.6.2 Coarse grid occupancy

The canonical canvas is partitioned into a 3×3 grid. Normalized foreground counts form $h_g(M)$. Unlike the centroid-normalized polar term, this

component retains coarse location and balance on the writing canvas.

3.6.3 Projection profiles

Normalized row sums $h_y(M)$ and column sums $h_x(M)$ describe vertical and horizontal mass allocation. Their similarities are averaged.

For distributions p and q , with $m = (p + q)/2$, we use

$$D_{JS}(p, q) = \frac{1}{2}D_{KL}(p||m) + \frac{1}{2}D_{KL}(q||m),$$

with base-2 logarithms, and define

$$JSS(p, q) = 1 - \sqrt{D_{JS}(p, q)}.$$

Because $D_{JS} \in [0, 1]$, $JSS \in [0, 1]$. The component similarities are

$$\begin{aligned} s_p &= JSS(h_p(U), h_p(R)), \\ s_g &= JSS(h_g(U), h_g(R)), \\ s_{proj} &= \frac{1}{2} [JSS(h_x(U), h_x(R)) \\ &\quad + JSS(h_y(U), h_y(R))]. \end{aligned}$$

The frozen development specification is

$$S_{ASDS} = 100(0.70s_p + 0.15s_g + 0.15s_{proj}).$$

The development search placed 0.70 weight on the polar term. Rather than retain the more sample-specific unequal secondary weights, we split the remaining 0.30 equally between grid and projection evidence before freezing the reported specification.

The score does not use character ID, writer ID, style ID, or a human rating at inference. Nevertheless, its feature choice and weights were finalized after observing the original 150-pair study. It is therefore a development-stage algorithm with resampling-based internal validation rather than an independently confirmed score. Features, bins, weights, alignment, candidate search, and selection policy are reported so the remaining risk of retrospective optimism is explicit.

ASDS uses only the union of the five direction masks. Consequently, a six-channel parser is not mathematically necessary merely to construct the binary silhouette consumed by ASDS in the present direct-digital modality. ASDS is

one scalar consumer of the parsing pipeline, rather than the sole justification for it. The separate channels remain necessary for the deployed production score, per-direction agreement, missing/extra directional evidence, endpoint relations, and localized feedback. We therefore include a paired direct-ink ASDS ablation that applies the same alignment, descriptors, and frozen weights directly to thresholded source ink. This ablation isolates whether parsing supplies any measurable normalization benefit to the scalar score; it does not test the channel-level outputs that require parsing.

3.7 Evidence extraction and feedback

The system preserves component scores, selected transform, pre-alignment centre distance, ink-area ratio, per-direction agreement, and keypoint agreement. Difference masks are summarized by direction channel, missing reference ink versus extra learner ink, and a 3×3 region set. Connected components of the five direction masks are exposed as direction-classified regions, not as recovered stroke order.

Deterministic feedback ranks centre, scale, keypoint, and local-direction findings. An optional text model receives only this structured contract and may improve fluency; it cannot change the score or invent unsupported brushwork or stroke-order claims.

4 Experimental Protocol

4.1 Legacy segmentation corpus and quality control

The recovered legacy corpus contains 894 sample directories across 43 character IDs. Complete RGB images, five direction masks, a keypoint mask, and a stacked six-channel mask are available for 840 samples from character IDs 0–39. Fifty-four samples from IDs 40–42 lack the character-specific direction mapping and are excluded rather than pseudo-labelled.

File completeness alone was insufficient. A frozen semantic QC procedure identified 12 source-image/GT mismatches and 59 non-canonical members of exact image-and-mask

duplicate groups. The exclusion sets do not overlap, leaving 769 unique QC-clean observations. Duplicate-group membership and mismatch decisions were fixed before the formal model comparison. No exact duplicate group crosses the original train/validation/test assignment.

The original standard assignment contains 600/120/120 samples and becomes 530/119/120 after QC. A second assignment was frozen before evaluation with 28/6/6 disjoint character identities and originally 588/126/126 samples; after the same exclusions it contains 539/114/116 train/validation/test observations. The source assignment, derived QC-clean split, and exclusion contract are content-hashed; their complete SHA-256 values are given in the reproducibility supplement. Writer identity cannot be recovered and writer-disjoint performance is not claimed.

4.2 Training and segmentation evaluation

We compare a 32-base-channel U-Net, ResNet-50 DeepLabV3+, and ImageNet-pretrained SegFormer-B2. Each architecture is run with seeds 20260811, 314159, and 271828 on both split protocols, for 18 formal runs. Inputs are 512×512 . Batch size is 8 for U-Net and 4 for the two larger models. Label-safe augmentation includes translation up to 12 pixels, isotropic scale 0.9–1.1, brightness 0.85–1.15, contrast 0.9–1.1, and Gaussian blur with probability 0.15.

All models use AdamW, weight decay 0.01, automatic mixed precision, gradient clipping at 1.0, three warm-up epochs, cosine decay to 10^{-6} , at most 120 epochs, and early stopping with patience 15. U-Net uses learning rate 3×10^{-4} . DeepLabV3+ uses encoder/decoder rates $10^{-4}/10^{-3}$; SegFormer-B2 uses $3 \times 10^{-5}/3 \times 10^{-4}$. Positive BCE weights are estimated from at most 200 training samples and capped at 100. Thresholds are calibrated using the validation set only.

We report five-direction Macro Dice and Macro IoU, per-channel precision/recall/Dice, Boundary F1, strict keypoint F1, and tolerant keypoint F1 at one, three, and five pixels. Test results are summarized as the mean and sample standard deviation across all declared seeds.

4.3 Reference library

The structural experiments use 200 approved references: 100 in the Calli-Tongji Beta Ouyang Xun regular-script category and 100 in the Wang Xizhi running-script category. The reference cache contains six binary masks generated by the frozen SegFormer-B2 v1 checkpoint. It supports seven same-character cross-style pairs but no same-style, different-instance pair. The restricted source images are not redistributed.

4.4 Controlled perturbation benchmark

To isolate scoring and registration from parser error, deterministic perturbations are applied directly to the cached real-reference masks. Nuisance families are translation, rotation, scale up, scale down, and a compound allowed transform. Structural families are terminal deletion, added fragment, local fragment shift, direction dilation, direction erosion, and keypoint shift. Target direction channels and movement axes are selected with stable hashes rather than model errors or manual case choice. A transformed case that risks canvas clipping is marked invalid with a reason and excluded from valid-score summaries.

For each family we report valid N , mean, median, sample standard deviation, 5th and 95th percentiles, 95% reference-bootstrap confidence intervals, invalid fraction, within-reference severity Spearman correlation, and adjacent monotonicity.

4.5 Paired registration ablation

The identical reference/perturbation/severity tuples are evaluated with: (i) no alignment; (ii) the production constrained alignment; and (iii) a preregistered wider search with $s \in [0.6, 1.4]$ and $\phi \in [-12^\circ, 12^\circ]$. Comparisons are paired. Confidence intervals resample references, while Wilcoxon signed-rank tests and rank-biserial effect sizes use per-reference mean paired differences. Family-level comparisons are the primary ablation; perturbation-level p values are unadjusted, descriptive breakdowns.

4.6 Score audit and cross-reference semantics

The score audit counts inactive direction channels and recomputes the coverage-aware candidate described in Sect. 3.5. Qualitative examples are selected by correction quantiles—maximum, upper quartile, median, lower quartile, and near-zero positive—rather than appearance.

For cross-reference semantics, all seven available same-character cross-style pairs and 50 score-independent different-character negative pairs are evaluated. Self-pairs are prohibited. The absent same-character, same-style/different-instance condition is reported as unsupported rather than synthesized.

4.7 Diagnostic benchmark

Perturbation cause supplies deterministic diagnostic ground truth. We report Recall@3, Top-1 cause, direction channel, missing/extra type, exact and overlap grid localization, specificity, and centre-direction wording. Language-model text is disabled. Because the benchmark describes engineered perturbations rather than expert pedagogy, it is a secondary analysis.

4.8 Original blinded human study

Before collecting ratings, 150 natural same-character, different-instance pairs were selected from the 769 QC-clean observations. All 40 character IDs are represented by three or four pairs. Bad images, exact duplicates, and suspected near duplicates were excluded. Pair roles, asset hashes, blinded identifiers, and display order were frozen. Scores and source identifiers were hidden from raters.

Three consenting adult raters independently judged structural similarity on a five-point ordinal scale. Their self-reported backgrounds were heterogeneous: one had one year of calligraphy learning/teaching experience, one reported two years of other relevant experience, and one reported none. We therefore use “human raters”, not “three experts”. Each rater saw 150 canonical pairs and 15 randomly interspersed hidden repeats.

Validity is measured by Spearman correlation between each score and the three-rater mean on canonical pairs. Confidence intervals

use 10,000 bootstrap resamples clustered by target character. Inter-rater reliability uses two-way random-effects, absolute-agreement ICC(2,1) and ICC(2, k). Hidden repeats are excluded from correlation and ICC and are used only for exact agreement, agreement within one point, mean absolute difference, and quadratic weighted kappa.

4.9 Post-rating ASDS development

The original 150 ratings constitute a development set for ASDS, not an independent test. Three fixed distribution families were considered: polar occupancy, 3×3 grid occupancy, and projection profiles. A coarse non-negative simplex search evaluated component weights. The published formula is deliberately regularized: the search-selected polar weight is retained, while the remaining weight is divided equally between grid and projection terms. The full-grid apparent optimum was $(w_g, w_p, w_{\text{proj}}) = (0.225, 0.700, 0.075)$ with $\rho = 0.557$; the frozen 0.15/0.70/0.15 formula gives $\rho = 0.556$. Five-fold cross-validation groups by character ID; each fold’s weights are selected on the other characters only. Character-cluster bootstrap intervals are reported for the full frozen formula, grouped out-of-fold predictions, and paired differences from the production and coverage-aware scores.

Component ablations report the polar, grid, and projection terms separately. The combined formula is compared with each component so that any apparent benefit cannot be attributed to an unreported learned regressor. Development component comparisons are exploratory and are not multiplicity-adjusted.

4.10 Paired direct-ink ASDS ablation

To test whether ASDS itself benefits from the six-channel parser, we recompute the frozen score on the same 150 rated pairs using foreground extracted directly from each original direct-digital RGB raster. Images are converted to grayscale and foreground is defined by intensity < 240 , the threshold already frozen for dataset QC. No learned preprocessing is used. The direct-ink masks are passed through the same constrained

alignment and the same 0.70/0.15/0.15 ASDS formula; neither the threshold, alignment policy, features, nor weights are selected against the ratings for this comparison.

We report Spearman association with the three-rater mean for parsed-union and direct-ink ASDS, their paired difference, and their mutual rank association. Uncertainty is estimated with 10,000 paired bootstrap resamples clustered by character ID. Because the ablation uses the original ASDS development pairs, it assesses implementation dependence within the same retrospective development set rather than external validity.

5 Results

5.1 RQ1: segmentation and unseen-character generalization

Table 1 shows that, on the QC-clean standard split, DeepLabV3+ obtains the highest direction Macro Dice (0.9630 ± 0.0006), Macro IoU (0.9287 ± 0.0012), and Boundary F1 (0.8443 ± 0.0028). SegFormer-B2 is close in direction overlap but shows greater between-seed variation. U-Net is weaker on direction and boundary metrics yet gives the best strict keypoint F1 (0.7562 ± 0.0038). Thus, dense direction segmentation and sparse keypoint localization favour different architectures.

All models degrade when character identities are held out (Table 2). SegFormer-B2 gives the strongest character-disjoint direction result: Macro Dice 0.8866 ± 0.0008 , Macro IoU 0.7977 ± 0.0013 , and Boundary F1 0.6851 ± 0.0046 . Its Macro-Dice decrease from the standard split is 7.150 percentage points, versus 9.664 for DeepLabV3+ and 9.360 for U-Net. DeepLabV3+ remains second for direction overlap, while U-Net retains the highest strict keypoint F1. SegFormer-B2 is therefore the strongest evaluated model for unseen-character direction parsing, not a universal winner across all six outputs.

The earlier single SegFormer release achieved direction Macro Dice 0.9610 but was trained before semantic QC. No exact duplicate group crosses its standard split, so its result is not explained by exact train-test duplication. Nevertheless, its training population included 12 mismatched and duplicate-weighted observations; all

formal comparisons above use the frozen 769-sample QC-clean population.

5.2 RQ2: controlled score behaviour and registration

The family-level results in Table 3 show heterogeneous residual nuisance sensitivity. Translation is recovered exactly in all 800 valid observations. Mean absolute penalties are 5.395 for rotation, 13.653 for scale-up, and 16.356 for scale-down. Scale-up and compound transformations have invalid fractions of 49.0% and 54.0%, respectively, because transformed support risks leaving the canvas; those cases are reported rather than silently counted as score failures. Scale-related summaries are therefore conditional on clipping-safe references. Because clipping safety is related to glyph extent, these conditional means should not be interpreted as full-library invariance.

Structural perturbations generally lower the score with severity. Direction dilation and erosion have the largest average penalties, followed by terminal deletion and added fragments. Local fragment shifts and keypoint shifts are penalized more weakly. The benchmark therefore identifies both a desired signal—sensitivity to local structure—and a principal weakness—residual scale sensitivity. Full per-family distributional statistics are provided in Supplementary Table S1.

The paired comparison in Table 4 shows that, relative to no alignment, the constrained method reduces nuisance penalties by 30.118 points (reference-bootstrap 95% CI 29.506–30.727) and preserves an additional 11.189 points of structural penalty (10.094–12.322). Both per-reference signed-rank comparisons are nonzero with rank-biserial effect size 1.0.

The wider alignment does not dominate the production range. The current range has a 2.896-point lower nuisance penalty (2.757–3.034), whereas the wider range preserves 0.284 additional points of structural penalty. A wider search therefore fails to reduce residual nuisance sensitivity and provides only a small structural-sensitivity advantage. The paired family-level breakdown is reported in Supplementary Table S2.

Table 1 QC-clean standard-split segmentation results over three fixed random seeds. Values are mean $\pm$ sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.9196 $\pm$ 0.0018	0.8515 $\pm$ 0.0030	0.7562 $\pm$ 0.0038	0.8009 $\pm$ 0.0023
DeepLabV3+	0.9630 $\pm$ 0.0006	0.9287 $\pm$ 0.0012	0.7218 $\pm$ 0.0012	0.8443 $\pm$ 0.0028
SegFormer-B2	0.9581 $\pm$ 0.0031	0.9196 $\pm$ 0.0057	0.7139 $\pm$ 0.0046	0.8259 $\pm$ 0.0088

Table 2 Character-disjoint test results over three fixed random seeds. Train, validation, and test character identities are disjoint. Values are mean $\pm$ sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.8260 $\pm$ 0.0161	0.7120 $\pm$ 0.0225	0.6972 $\pm$ 0.0045	0.6430 $\pm$ 0.0136
DeepLabV3+	0.8664 $\pm$ 0.0094	0.7663 $\pm$ 0.0146	0.6755 $\pm$ 0.0031	0.6728 $\pm$ 0.0138
SegFormer-B2	0.8866 $\pm$ 0.0008	0.7977 $\pm$ 0.0013	0.6604 $\pm$ 0.0033	0.6851 $\pm$ 0.0046

5.3 Score semantics: inactive channels and cross-reference tests

Table 5 shows that 41 of 200 references contain an inactive direction channel: 36 contain one and five contain two; none contains three or more. Empty-channel agreement affects 1,252 valid perturbation rows. Replacing the all-five mean by an active-channel mean changes the score by -0.289 points on average and by as much as -15.402 points. The largest corrections occur when inactive channels receive perfect Dice credit while a perturbed active channel degrades.

Figure 3 illustrates the prespecified correction quantiles without selecting examples for visual appeal.

The seven same-character cross-style pairs in Table 6 have mean score 20.239 (bootstrap 95% CI 15.639–25.069), compared with 9.741 (8.584–10.970) for 50 different-character negatives. Cliff’s delta is 0.84 [6]. This is evidence of a same-character reference signal, but the small positive set and missing same-style/different-instance condition prevent a broad style-generalization claim.

5.4 RQ3: blinded human validity and ASDS development

Across 150 canonical pairs, S_{prod} has a positive but modest association with the three-rater mean ($\rho = 0.297$, character-cluster bootstrap 95% CI 0.148–0.441, $p = 0.00023$). The coverage-aware audit is stronger ($\rho = 0.429$, 0.313–0.535, $p < 10^{-7}$). The paired bootstrap difference is 0.132



Fig. 3 Coverage-correction cases selected by correction quantiles rather than visual preference. Purple denotes aligned overlap, blue missing-reference ink, and red extra-user ink.

(0.062–0.197). This result is consistent with the inactive-channel audit but does not, by itself, authorize post-hoc replacement of the production score.

These associations are summarized in Table 7. Canonical ICC(2,1) is 0.448 and ICC(2,k) is 0.709 (Table 8). Across 45 rater-level hidden repeats,

Table 3 Controlled perturbation summary. Nuisance rows report absolute score drop; structural rows report signed drop from identity. Full distribution and severity statistics are in Supplementary Table S1.

Perturbation	Valid N	Mean drop	95% CI	Invalid fraction	Monotonic
Translation	800	0.000	[0.000, 0.000]	0.000	1.000
Rotation	520	5.395	[5.079, 5.747]	0.133	0.647
Scale up	408	13.653	[12.964, 14.420]	0.490	0.271
Scale down	800	16.356	[15.670, 17.074]	0.000	0.500
Compound allowed transform	276	8.143	[7.621, 8.717]	0.540	0.466
Terminal deletion	800	19.757	[17.387, 22.087]	0.000	0.992
Added direction fragment	800	11.398	[9.934, 12.929]	0.000	0.987
Local fragment shift	800	4.310	[3.886, 4.772]	0.000	0.985
Direction dilation	800	24.034	[22.180, 25.955]	0.000	1.000
Direction erosion	800	22.225	[20.269, 24.344]	0.000	0.997
Keypoint shift	800	7.468	[7.294, 7.641]	0.000	1.000

Table 4 Paired alignment comparison. Positive benefit differences favour the current constrained alignment. Wilcoxon tests use per-reference mean paired differences.

Comparison	Family	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	nuisance	200	8.455	38.573	[29.506, 30.727]	< 0.001	1.000
Ours–none	structural	200	14.865	3.677	[10.094, 12.322]	< 0.001	1.000
Ours–wide	nuisance	200	8.455	11.351	[2.757, 3.034]	< 0.001	1.000
Ours–wide	structural	200	14.865	15.149	[-0.355, -0.213]	< 0.001	-0.683

Table 5 Inactive direction channels among 200 references.

Inactive channels	References	Fraction
0	159	0.795
1	36	0.180
2	5	0.025

exact agreement is 48.9%, agreement within one scale point is 93.3%, mean absolute difference is 0.578, and pooled quadratic weighted kappa is 0.634. Individual production-score correlations are positive for all raters and range from 0.174 to 0.350. Average ratings are therefore more reliable than a single rating, but meaningful subjectivity remains.

ASDS raises the development-set association to $\rho = 0.556$ (character-cluster bootstrap 95% CI 0.454–0.651). Predictions produced by character-grouped five-fold weight selection retain $\rho = 0.540$ (0.432–0.639). The paired bootstrap interval for ASDS minus production is 0.110–0.410, and that for ASDS minus the coverage-aware audit is 0.017–0.239. For grouped out-of-fold predictions, the difference from coverage-aware scoring is less certain (−0.003–0.224).

The component analysis in Table 9 identifies the polar term as the strongest individual spatial descriptor ($\rho = 0.547$); grid and projection similarities reach 0.447 and 0.445. Thus, most ASDS association comes from radial/angular ink organization, while the two coarse canvas terms provide a small complementary contribution. The combined score is not claimed to be significantly better than the polar term alone.

These results show that human ratings are more closely related to spatial ink organization than to the deployed channel/pixel overlap alone. They are not an external validation of ASDS, because the same 150 ratings informed its feature design. The grouped out-of-fold estimate reduces weight-selection optimism and is reported as internal validation; the full-sample value remains descriptive and retrospective.

The paired direct-ink ablation (Table 10) gives $\rho = 0.55623$ for parsed-union ASDS and $\rho = 0.55583$ for direct-ink ASDS. Their difference is 0.00040, with character-cluster bootstrap 95% CI −0.01011–0.01030. The two score vectors are themselves strongly rank-associated ($\rho = 0.9973$). Thus, on these clean direct-digital inputs, the parser neither measurably improves nor degrades the scalar ASDS association. This result narrows the claim: ASDS is a silhouette score, whereas the

Table 6 Cross-reference structural agreement.

Pair type	N	Mean	Median	Mean 95% CI
Same character, cross style	7	20.239	20.272	[15.639, 25.069]
Different-character negative	50	9.741	8.869	[8.584, 10.970]

Table 7 Association between the structural scores and the mean rating of three blinded human raters across 150 canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by character.

Score	ρ	95% CI	p
Production	0.297	[0.148, 0.441]	2.28×10^{-4}
Coverage-aware	0.429	[0.313, 0.535]	4.32×10^{-8}

Table 8 Human-rater reliability. ICC uses only the 150 canonical presentations. Repeat statistics use the 15 hidden repeats per rater.

Rater	Exact	Within ± 1	MAD	QWK
E01	0.600	1.000	0.400	0.559
E02	0.467	1.000	0.533	0.721
E03	0.400	0.800	0.800	0.591
POOLED	0.489	0.933	0.578	0.634

Canonical ICC(2,1)=0.448 and ICC(2,k)=0.709 for 150 pairs and 3 raters.

parser is justified by the separate direction, overlap, and endpoint evidence evaluated in RQ1 and returned by the application.

Figure 4 visualizes the stronger monotonic organization of ASDS while preserving the substantial dispersion that motivates replication on independently collected writers and characters.

5.5 RQ4: diagnostic feedback

Table 11 separates the dominant localization failure types. Required Recall@3 is 0.701, whereas exact grid-region localization is 0.109. Among exact-region failures, 53.9% occur because the perturbation truth spans multiple cells while the metric requires one exact cell, 24.6% involve an incorrect direction channel, and 20.1% contain no local finding in the top three. Wrong missing/extra type and residual registration shifts are uncommon. The main limitation is therefore a

mixture of localization representation and direction attribution rather than wording. This motivates a future set- or pixel-overlap localization metric, not further tuning of feedback rules on the same cases.

6 Discussion

6.1 What the experiments establish

The first result is that model selection depends on the generalization target. DeepLabV3+ is strongest on the QC-clean standard split, but SegFormer-B2 has the best character-disjoint direction overlap and the smallest decline between protocols. U-Net remains strongest on strict key-point localization. This architecture-by-output interaction argues against selecting a parser from one aggregate metric. It also explains why the preliminary standard-split SegFormer result was insufficient evidence for unseen-character performance.

The second result is that restricted registration is not merely an implementation detail. Without alignment, the tested global geometric changes dominate the score. Relative to no alignment, the production range substantially reduces nuisance sensitivity while retaining structural penalties. The wider range does not improve this trade-off, providing a useful negative result: more transform capacity is not automatically better for diagnostic comparison. Scale remains the main weakness. Moreover, scale-up and compound summaries are conditional on the clipping-safe subset, with invalid fractions of 49.0% and 54.0%. Because clipping safety depends on glyph extent, these results should not be interpreted as invariance over the full reference library. Further adjustment on the same perturbation suite would also risk benchmark-specific overfitting.

The third result is semantic. Both-empty direction channels can contribute perfect Dice even though no active structure is compared. The average effect is small, but a limited set of cases receives a large correction. The stronger human

Table 9 Development-stage ASDS component ablation and association with mean human structural rating. Confidence intervals use 10,000 character-cluster bootstrap resamples. ASDS was developed after the original ratings and is not an independent result.

Score or component	Spearman ρ	95% CI
Grid JS similarity only	0.447	[0.303, 0.572]
Projection JS similarity only	0.445	[0.316, 0.564]
Polar JS similarity only	0.547	[0.443, 0.645]
ASDS frozen development formula	0.556	[0.454, 0.651]
ASDS character-grouped OOF	0.540	[0.432, 0.639]

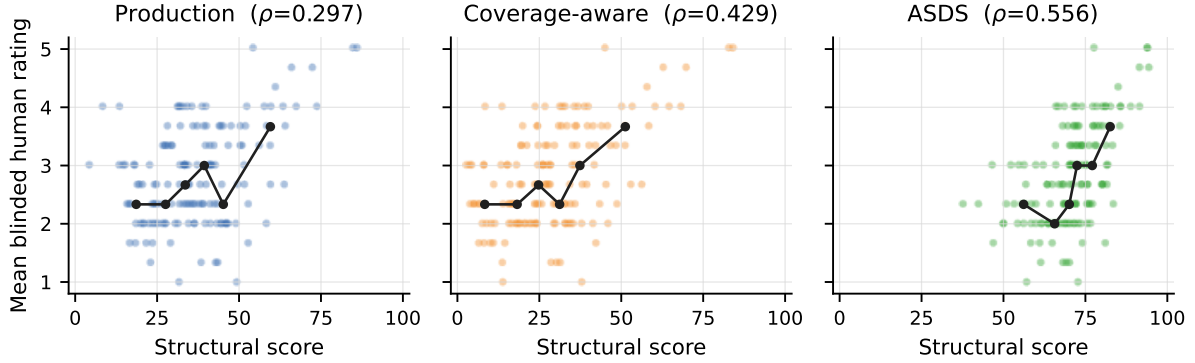


Fig. 4 Development-stage association between blinded mean structural rating and the production, coverage-aware, and ASDS scores for 150 natural same-character pairs. Points are individual pairs; lines connect score-bin medians only as a descriptive guide. ASDS was designed after these ratings, so this figure is not an independent validation.

Table 10 Direct-ink ablation of the frozen ASDS specification on the same 150 development pairs.

ASDS input	Spearman ρ	95% CI
Parsed union	0.556	[0.454, 0.651]
Direct ink	0.556	[0.454, 0.649]
$\Delta\rho$ (parsed-direct)	0.000	[-0.010, 0.010]

Both variants use the same alignment policy, ASDS features, and frozen weights. Direct ink uses the frozen dataset-QC rule: grayscale intensity < 240. No threshold or weight was selected against the ratings.

Table 11 Primary taxonomy of exact-region diagnostic failures.

Failure type	Count	Fraction
Multi-region/boundary truth	1922	0.539
Wrong direction channel	876	0.246
Local finding absent from Top-3	718	0.201
Wrong missing/extra type	38	0.011
Alignment residual shift	11	0.003

association of the coverage-aware audit supports this concern. ASDS extends the finding: coarse radial, angular, and projection distributions track the human structural rating better than the production score in the development study. This is plausible because raters can perceive global proportion and mass distribution even when exact channel overlap is modest.

6.2 Retrospective development and internal validation

It would be statistically inappropriate to present $\rho = 0.556$ as a final generalization result. The spatial representation and weights were finalized after the original ratings were available. Character-grouped out-of-fold predictions reduce weight-selection optimism, but they cannot erase the fact that feature families were proposed after observing the same study.

We therefore preserve two distinct layers of evidence: (i) the originally frozen production and coverage-aware results, which were not selected to

maximize the observed ASDS association; and (ii) a transparent post-rating development analysis including all considered feature families, component ablations, character-cluster uncertainty, and character-grouped out-of-fold weight selection. The latter is an internal-validation estimate, not evidence from newly collected material.

ASDS is also intentionally small. Its three bounded components are directly inspectable and can be computed on CPU. Component ablations test whether the combined score adds value beyond its dominant polar term. The complete development record is retained to make researcher degrees of freedom visible. External replication should keep this specification fixed rather than retune it on the next rating set.

6.3 Why retain the six-channel parser when ASDS uses union ink?

ASDS is a spatial silhouette descriptor: polar occupancy, 3×3 occupancy, and projection profiles are all computed from the union of the five direction masks. In a direct-digital-writing application with an already clean binary raster, the parser is therefore not mathematically required merely to obtain the silhouette used by ASDS. Claiming otherwise would overstate the role of parsing.

The parser serves a broader evidential contract. The deployed production score uses channel-wise Dice and endpoint agreement; the application reports which direction family is missing or excessive, where the discrepancy lies, and whether stroke-endpoint relations disagree. These outputs cannot be recovered from a union silhouette when a direction-label swap leaves the ink support unchanged. ASDS should therefore be interpreted as one scalar consumer that complements, rather than replaces, the channel-level evidence. The paired ablation answers the scalar question directly: parsed-union and direct-ink ASDS differ by only 0.0004 in Spearman correlation, and the paired 95% interval includes small advantages in either direction. We therefore find no evidence that parsing improves ASDS through denoising or standardization on the present clean direct-digital inputs. This negative result is informative rather than adverse: it prevents the scalar score from being used as circular justification for the parser. The six-channel representation remains

necessary for the direction-specific and endpoint-specific diagnostic claims evaluated in RQ1 and delivered by the system, but it should not be required when an application requests only ASDS from an already clean binary raster.

6.4 Interpretation of human agreement

The average-rating ICC is substantially higher than the single-rating ICC. This supports averaging multiple judgments for the validation endpoint, while also showing that the task is not objective in the sense of a segmentation mask. The rater pool is small and heterogeneous. One rater had limited calligraphy teaching/learning experience, but the group is not a panel of senior calligraphy experts. The results justify the phrase “human-audited structural agreement”; they do not justify “expert aesthetic score”.

A stronger follow-up should recruit at least two experienced calligraphy educators, record background through a fixed questionnaire, and analyse whether professional and non-professional rankings differ. Such a study would extend the construct from general structural similarity toward pedagogical quality, but it should not be retroactively substituted for the current task.

6.5 Diagnostic and application implications

The low exact-region result initially appears to indicate poor feedback. Failure analysis shows that more than half of these errors arise from a single-cell metric applied to multi-cell perturbations. The next diagnostic benchmark should store affected support as a region set or pixel mask and use overlap-based scoring. Wrong direction attribution remains a genuine model or evidence-extraction failure and warrants separate study. Optional language generation should be evaluated only after the structured evidence contract is validated; fluent text cannot repair an incorrect direction or region.

The target platform receives direct digital writing. This modality is aligned with the held-out legacy images, but it does not imply robustness to phone photographs, paper texture, illumination, or perspective. The SegFormer-B2 parser benefits from a GPU for interactive latency; reference

masks are precomputed, and registration, ASDS, deterministic feedback, and language-API calls do not require a dedicated training-class GPU. Distillation and quantized edge deployment are engineering extensions rather than claims of this study.

6.6 Limitations and threats to validity

The corpus contains only 40 fully labelled character identities. Writer identity and verified style provenance are unavailable, so character-disjoint evaluation is not writer-disjoint evaluation. The standard and character-disjoint results come from one recovered data source, not an external corpus. Exact duplicate leakage has been audited, but unobserved collection-specific regularities may remain.

The structural perturbation experiments operate on cached reference masks and isolate score behaviour from parser errors. This is a deliberate internal validity choice, but an end-to-end error can combine segmentation, registration, and scoring failures. The reference library contains only two style categories, seven same-character cross-style pairs, and no same-style different-instance pairs. Broader course coverage cannot be inferred from these comparisons.

The human study contains 150 pairs and three raters from the same recovered corpus. Its character-grouped analysis limits leakage across character identities but does not establish generalization to new writers, collections, or independently recruited evaluator populations. Finally, ASDS uses the union of five direction masks and can ignore a direction-label swap that leaves overall ink distribution unchanged. It should therefore complement, not erase, the channel-level evidence returned to users.

7 Conclusion

We presented a reference-conditioned pipeline for low-resource Chinese calligraphy analysis that combines overlapping direction/keypoint parsing, restricted global registration, interpretable structural comparison, and localized evidence. Across three fixed seeds, DeepLabV3+ is strongest on the QC-clean standard split, while SegFormer-B2 gives the best character-disjoint direction result

and U-Net gives the best strict keypoint result. Controlled perturbations and paired ablations show that the constrained registration substantially reduces nuisance sensitivity relative to no alignment without freely warping away local structural changes. Scale-related results remain conditional on clipping-safe references and do not establish full-library transform invariance.

The human study also changes the interpretation of the score. The deployed formula has a modest positive association with blinded structural-similarity ratings, and a coverage-aware audit is stronger. The proposed ASDS score, which compares polar occupancy, coarse grid occupancy, and projection profiles, raises development-stage Spearman correlation from 0.297 to 0.556 and retains 0.540 under character-grouped out-of-fold prediction. Because ASDS was developed after observing the original ratings, these values are retrospective internal-validation evidence rather than external confirmation. They support ASDS as a compact candidate score while leaving replication on independently collected material as future work.

A paired direct-ink ablation further shows that parsed-union and raw-ink ASDS are nearly equivalent on the clean direct-digital study inputs ($\Delta\rho = 0.0004$, 95% CI -0.0101 – 0.0103). The scalar spatial descriptor therefore does not itself require the parser in this modality. The parser’s independent value lies in overlapping direction segmentation, endpoint evidence, the deployed channel-aware score, and localized missing/extra diagnostics.

The resulting system should be viewed as an auditable structural-analysis tool, not a universal aesthetic grader. Its strongest scientific contribution is the joint evaluation discipline: quality-controlled and character-disjoint parsing, controlled registration tests, semantic score audits, blinded human association, explicit post-hoc development, and a character-grouped internal-validation analysis.

Acknowledgements

The authors thank Chengcheng Wan for project supervision and the adult volunteers who contributed blinded structural-similarity ratings.

Declarations

Funding

This work was conducted within the East China Normal University Undergraduate Innovation Training Incubation Project (Entrepreneurship Training category), “OneStroke: A Vision-Model-Based Intelligent Feedback System for Chinese Calligraphy”. The project application records no external funding channel; no grant number was available at the time of writing.

Conflict of interest

The authors declare no competing interests.

Ethics approval and consent to participate

The structural-similarity study involved adult volunteers, stored ratings under evaluator codes, and did not collect sensitive personal information. Before submission, the authors will insert the institution-specific ethics approval, exemption, or review statement and the applicable consent wording. No approval or exemption is claimed in this draft.

Consent for publication

Not applicable, subject to confirmation by the authors before submission.

Data availability

The recovered segmentation corpus and the Calli-Tongji-derived reference images are subject to their original ownership and licence constraints. Reproducibility artifacts provide frozen identifiers, hashes, split contracts, statistics, and generation scripts without redistributing restricted source images. The exact public/private release statement will be supplied before submission.

Code availability

Source code, frozen experiment contracts, and non-restricted reproducibility artifacts are available at <https://github.com/LiuXiaofan-0321/OneStroke2026>. If anonymous review is required, an identity-free archival copy will be supplied through the submission system.

Author contributions

X.L.: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, project administration. R.Z.: Data curation, methodology, software, validation, writing—review and editing. Y.F.: Software, system integration, validation, writing—review and editing. All authors contributed to the investigation, approved the submitted manuscript, and accept responsibility for the work.

References

- [1] Baloun J, Prantl M, Lenc L, et al (2025) On self-supervision in historical handwritten document segmentation. *International Journal on Document Analysis and Recognition* 28(3):329–344. <https://doi.org/10.1007/s10032-025-00538-6>
- [2] Belongie S, Malik J, Puzicha J (2002) Shape matching and object recognition using shape contexts. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 24(4):509–522. <https://doi.org/10.1109/34.993558>
- [3] Chen LC, Zhu Y, Papandreou G, et al (2018) Encoder-decoder with atrous separable convolution for semantic image segmentation. In: *European Conference on Computer Vision*, pp 833–851, https://doi.org/10.1007/978-3-030-01234-2_49
- [4] Chen W, Su J, Song W, et al (2024) Quality evaluation methods of handwritten chinese characters: A comprehensive survey. *Multimedia Systems* 30(4):194. <https://doi.org/10.1007/s00530-024-01396-8>
- [5] Chen W, Zhu G, Li Y, et al (2025) Siamhcc: A novel siamese network for quality evaluation of handwritten chinese characters. *Multimedia Systems* 31(4):303. <https://doi.org/10.1007/s00530-025-01886-3>
- [6] Cliff N (1993) Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin* 114(3):494–509. <https://doi.org/10.1037/0033-2909.114.3.494>

- [7] Efron B, Tibshirani RJ (1993) An Introduction to the Bootstrap. Chapman & Hall/CRC, New York, <https://doi.org/10.1201/9780429246593>
- [8] Gong X, Bai Z, Nie H, et al (2024) Stroke-seg: A deep learning-based framework for chinese stroke segmentation. IET Image Processing 18(13):4341–4355. <https://doi.org/10.1049/ipr2.13255>
- [9] Jaderberg M, Simonyan K, Zisserman A, et al (2015) Spatial transformer networks. In: Advances in Neural Information Processing Systems
- [10] Koo TK, Li MY (2016) A guideline of selecting and reporting intraclass correlation coefficients for reliability research. Journal of Chiropractic Medicine 15(2):155–163. <https://doi.org/10.1016/j.jcm.2016.02.012>
- [11] Lazebnik S, Schmid C, Ponce J (2006) Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. In: IEEE Computer Society Conference on Computer Vision and Pattern Recognition, pp 2169–2178, <https://doi.org/10.1109/CVPR.2006.68>
- [12] Lin J (1991) Divergence measures based on the shannon entropy. IEEE Transactions on Information Theory 37(1):145–151. <https://doi.org/10.1109/18.61115>
- [13] Liu CL, Yin F, Wang DH, et al (2013) Online and offline handwritten chinese character recognition: Benchmarking on new databases. Pattern Recognition 46(1):155–162. <https://doi.org/10.1016/j.patcog.2012.06.021>
- [14] Ma Z, Su J (2017) Aesthetics evaluation for robotic chinese calligraphy. IEEE Transactions on Cognitive and Developmental Systems 9(1):80–90. <https://doi.org/10.1109/TCDS.2016.2645598>
- [15] McGraw KO, Wong SP (1996) Forming inferences about some intraclass correlation coefficients. Psychological Methods 1(1):30–46. <https://doi.org/10.1037/1082-989X.1.1.30>
- [16] Oliveira SA, Seguin B, Kaplan F (2018) dhsegment: A generic deep-learning approach for document segmentation. In: 16th International Conference on Frontiers in Handwriting Recognition, pp 7–12, <https://doi.org/10.1109/ICFHR-2018.2018.00011>
- [17] Pengcheng G, Gang G, Jiangqin W, et al (2017) Chinese calligraphic style representation for recognition. International Journal on Document Analysis and Recognition 20(1):59–68. <https://doi.org/10.1007/s10032-016-0277-z>
- [18] Ronneberger O, Fischer P, Brox T (2015) U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention, pp 234–241, https://doi.org/10.1007/978-3-319-24574-4_28
- [19] Wang L, Gong X, Zhang Y, et al (2018) Artistic features extraction from chinese calligraphy works via regional guided filter with reference image. Multimedia Tools and Applications 77(3):2973–2990. <https://doi.org/10.1007/s11042-017-4688-8>
- [20] Wang M, Lu S, Hu Z, et al (2025) Overview of ccl25-eval task 11: Evaluation of the quality of handwritten chinese characters. In: Proceedings of the 24th China National Conference on Computational Linguistics. Chinese Information Processing Society of China, pp 452–460, URL <https://aclanthology.org/2025.ccl-2.54/>
- [21] Wilcoxon F (1945) Individual comparisons by ranking methods. Biometrics Bulletin 1(6):80–83. <https://doi.org/10.2307/3001968>
- [22] Xia W, Jin L (2009) A kai style calligraphic beautification method for handwriting chinese character. In: 10th International Conference on Document Analysis and Recognition, pp 798–802, <https://doi.org/10.1109/ICDAR.2009.59>
- [23] Xie E, Wang W, Yu Z, et al (2021) Segformer: Simple and efficient design for semantic segmentation with transformers. In: Advances in Neural Information Processing Systems, pp

- [24] Xu T, Wang K, Chen Z, et al (2026) Unicalli: A unified diffusion framework for column-level generation and recognition of chinese calligraphy. In: International Conference on Learning Representations, URL <https://openreview.net/forum?id=OSIPdrw56X>, arXiv:2510.13745
- [25] Zhang W, Ma H, Liu L, et al (2026) Callinet: A triplet network for chinese calligraphy style classification. International Journal on Document Analysis and Recognition 29(2):381–396. <https://doi.org/10.1007/s10032-025-00559-1>
- [26] Zhang X, Nagy G (2015) Computational method for calligraphic style representation and classification. Journal of Electronic Imaging 24(5):053003. <https://doi.org/10.1117/1.JEI.24.5.053003>
- [27] Zitová B, Flusser J (2003) Image registration methods: A survey. Image and Vision Computing 21(11):977–1000. [https://doi.org/10.1016/S0262-8856\(03\)00137-9](https://doi.org/10.1016/S0262-8856(03)00137-9)